Strategic Architecture for a German Waste Management Knowledge Graph: Packaging Codes, Material Identification, and Disposal Logic

1. Executive Introduction: The Semantic Gap in Waste Management

The construction of a high-fidelity knowledge graph for the German waste management sector represents a complex exercise in semantic alignment. The core challenge lies in bridging the gap between the static, theoretical identity of a packaging object—defined by its chemical composition and manufacturing codes—and its dynamic, context-dependent disposal pathway within the German *Kreislaufwirtschaft* (Circular Economy). A packaging engineer views a polyethylene terephthalate tray as a polymer with specific barrier properties (Code 01), while a waste sorting facility views it as a Near-Infrared (NIR) detectable object with a specific density profile. Simultaneously, a consumer in Berlin views it as an item for the "Wertstofftonne," whereas a consumer in a rural Bavarian district might be restricted to a "Gelber Sack" system or a communal depot.

To populate a knowledge graph that effectively models these interactions, one must synthesize data from three distinct strata: the regulatory frameworks that mandate identification (EU and National Law), the technical standards that define the semiotic codes (DIN and ISO), and the operational datasets that dictate local disposal rules (Municipal Bylaws and Dual System specifications). This report provides an exhaustive analysis of these data sources, offering a blueprint for a graph that does not merely list codes but encodes the logic of recyclability.

The urgency of such a system is underscored by the evolving regulatory landscape. The European Packaging and Packaging Waste Regulation (PPWR), currently in transition from the older Directive 94/62/EC, is shifting the industry from voluntary guidelines toward mandatory recyclability assessments.[1] In Germany, the implementation of the *Verpackungsgesetz* (VerpackG) has centralized authority in the *Stiftung Zentrale Stelle Verpackungsregister* (ZSVR), creating a definitive "Minimum Standard" for recyclability that the knowledge graph must reflect.[3] This report aggregates these disparate sources into a cohesive data dictionary suitable for immediate ingestion into a graph database.

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2. The Regulatory Ontology: From EU Directives to National Law

A robust knowledge graph must first establish the legal hierarchy that governs packaging identification. The codes found on packaging are not arbitrary; they are the result of specific legislative decisions. The graph's ontology must represent the relationship between the code (the signifier) and the legal authority (the signified regulation).

2.1 The Foundational Authority: Decision 97/129/EC

The root node for all material identification data in the European context is **Commission Decision 97/129/EC** of January 28, 1997. This decision established the alphanumeric system that remains the industry standard today.[5] While the overarching Directive 94/62/EC set the targets for packaging waste recovery, Decision 97/129/EC provided the technical nomenclature.

It is critical to note for the knowledge graph that, technically, the use of this marking system remains voluntary at the EU level, although the PPWR is moving toward making identification mandatory to facilitate consumer sorting.[1] However, the "voluntary" nature is legally nuanced: if a manufacturer *chooses* to mark the packaging to identify the material, they *must* use the numbering system defined in 97/129/EC. They cannot invent new codes for established materials.[5]

The graph must categorize these codes into specific material families defined by the Decision:

- **Plastics:** Codes 01 to 19.
- **Paper and Fiberboard:** Codes 20 to 39.
- **Metals:** Codes 40 to 49.
- **Wood:** Codes 50 to 59.
- **Textiles:** Codes 60 to 69.
- **Glass:** Codes 70 to 79.
- **Composites:** Codes 80 to 99.

This segmentation [7] forms the primary class structure of the ontology. Any instance of a packaging object in the graph must inherit properties from one of these parent classes.

2.2 National Transposition: The German Packaging Act (VerpackG)

In Germany, the EU directives are transposed into national law via the *Verpackungsgesetz* (VerpackG). For the knowledge graph, the VerpackG introduces the concept of **Systembeteiligungspflicht** (system participation obligation).[3] This is a boolean property for any packaging node: does this item accrue waste at the private consumer level? If Systembeteiligungspflicht = True, the manufacturer must license the packaging with a Dual

System (e.g., *Der Grüne Punkt*, *BellandVision*).[9]

The VerpackG also establishes the ZSVR as the central authority. The ZSVR's "Minimum Standard" (*Mindeststandard*) is the reference document for determining whether a specific material code actually translates to "recyclable" in practice.[4] For example, a packaging coded as "05 PP" (Polypropylene) is chemically recyclable, but if the ZSVR standard deems its specific form (e.g., a black tray with carbon pigments) as non-sortable, the knowledge graph must flag its Disposal_Outcome as "Thermal Recovery" rather than "Recycling".[12]

2.3 Technical Standardization: DIN 6120

While the EU Decision provides the *numbers*, the German standard **DIN 6120** provides the *syntax*. This standard, titled "Marking of Packaging Materials," dictates the graphical representation of the recycling triangle and the placement of the codes.[13]

A significant update to DIN 6120 in 2019 adjusted the rules for the "07 Other" category, aiming to reduce ambiguity by encouraging more specific identification of recyclates.[15] For a knowledge graph incorporating computer vision (e.g., scanning a package to identify its material), the geometry defined in DIN 6120 is the ground truth for training data. The standard ensures transparency and comparability, which are prerequisites for the automated sorting systems referenced in the ZSVR guidelines.[15]

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3. The Polymer Sub-Graph: Plastics Identification and Logic

Plastics represent the most complex subgraph due to the diversity of polymer types and the strict requirements for Near-Infrared (NIR) sorting in German facilities. The knowledge graph must map the consumer-facing codes to the industrial reality of polymer sorting.

3.1 Polyethylene Terephthalate (01 PET)

Code 01 is ubiquitous but operationally bifurcated in Germany. The graph must distinguish between the "Deposit Stream" (*Pfandsystem*) and the "Yellow Bin Stream" (*Gelber Sack*).

- **Material Definition:** A thermoplastic polymer resin of the polyester family.
- **Graph Logic:**
  - If Form = Beverage Bottle AND Volume < 3.0L: The destination is **Pfandautomat** (Deposit Return Machine). The material enters a high-purity recycling loop (bottle-to-bottle).[16]
  - If Form = Tray/Cup OR Content = Oil/Juice Shot/Non-Food: The destination is **Gelber Sack** (Yellow Sack). These items are sorted into the mixed PET fraction but generally do not re-enter the food-contact stream due to contamination risks.[17]

- **GS1 Mapping:** Maps to GDSN code POLYMER_PET.[19]

3.2 High-Density Polyethylene (02 PE-HD)

HDPE is a semi-crystalline thermoplastic known for its high strength-to-density ratio.

- **Typical Applications:** Detergent bottles, shampoo containers, milk jugs (though rare in Germany compared to Tetrapak).
- **Disposal Logic:** Highly recyclable via the **Yellow Bin**. German sorting plants use float-sink separation (Schwimm-Sink-Trennung); HDPE floats (density < 1 g/cm³), allowing easy separation from PET and PVC.[7]
- **Graph Attribute:** The graph should include a Floatability boolean attribute, set to True for Code 02.

3.3 Polyvinyl Chloride (03 PVC)

PVC is the "pariah" of the modern recycling stream. While it has a code, its presence is actively discouraged in packaging design guidelines.

- **Material Definition:** A rigid or flexible polymer containing chlorine.
- **Graph Warning:** PVC has a density similar to PET (~1.3-1.4 g/cm³), meaning it sinks in water separation tanks, contaminating the PET stream.
- **Disposal Route: Yellow Bin**, but it is typically sorted out as a contaminant or residue. It holds a negative value in the ZSVR recycling capability assessment if found in composite layers.[4]

3.4 Low-Density Polyethylene (04 PE-LD)

LDPE is defined by its flexibility and use in films.

- **Disposal Logic: Yellow Bin**.
- **Sorting Specifics:** Modern German sorting plants have specific "Film" (Folien) fractions. The graph must link Code 04 to the Fraction: Foil node. It is crucial that consumers separate LDPE lids from HDPE tubs to ensure the foil is blown into the correct chute by air separators.[8]

3.5 Polypropylene (05 PP)

PP is increasingly the preferred material for rigid packaging (tubs, pots, trays) due to its recyclability and lower density.

- **Disposal Logic: Yellow Bin**.
- **Graph Nuance:** The graph must handle the "Carbon Black" exception here. If an object is "05 PP" but colored with carbon black, it is NIR_Detectable = False and therefore Recyclable = False according to ZSVR, unless specific marker technologies are used.[12]

3.6 Polystyrene (06 PS)

- **Forms:** Rigid (Yoghurt pots) and Expanded (EPS/Styrofoam).
- **Disposal Logic:**
  - **Rigid PS: Yellow Bin**.
  - **EPS:** Small amounts in **Yellow Bin**; large amounts (e.g., appliance packaging) typically go to **Wertstoffhof** (Recycling Centers).[20]
- **Status:** PS is under pressure; many manufacturers are switching to PP or PET to improve recyclability metrics.

3.7 Other Plastics (07 O)

This is a catch-all category for Polycarbonate (PC), Polyamide (PA), Bioplastics (PLA), and complex laminates not covered by composite codes.

- **Graph Definition:** The "07" node represents high entropy and low recyclability.
- **Disposal: Yellow Bin**, but usually destined for "Mixed Plastics" leading to thermal recovery (incineration) rather than material recycling. Bioplastics marked 07 (e.g., PLA) are generally considered contaminants in the German recycling stream as they can compromise the quality of recycled PET or PE.[22]

Table 1: Plastic Data Dictionary for Knowledge Graph

Code	Abbreviation	Material Name (DE/EN)	Density (g/cm^3)	German Disposal	GS1 GDSN Code
01	**PET**	Polyethylenterephthalat	~1.38	Pfand (Bottle) / Gelber Sack (Tray)	POLYMER_PET
02	**PE-HD**	Polyethylen (Hart)	~0.95	Gelber Sack	POLYMER_HDPE
03	**PVC**	Polyvinylchlorid	~1.30-1.45	Gelber Sack (Residual)	POLYMER_PVC
04	**PE-LD**	Polyethylen (Weich)	~0.92	Gelber Sack (Foil)	POLYMER_LDPE

05	**PP**	Polypropylen	~0.90	Gelber Sack	POLYMER_PP
06	**PS**	Polystyrol	~1.05	Gelber Sack / Wertstoffhof	POLYMER_PS
07	**O**	Andere Kunststoffe	Var.	Gelber Sack (Thermal)	POLYMER_OTHER

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4. The Fiber and Metal Sub-Graphs: Codes 20-49

While plastics often dominate the recycling discussion, the Paper/Pasteboard/Cardboard (PPK) and Metal streams are statistically the most successfully recycled fractions in Germany. The knowledge graph must accurately model these materials, particularly the distinction between simple paper packaging and composites.

4.1 Paper and Cardboard (20-22 PAP)

The "Blaue Tonne" (Blue Bin) is the destination for these codes. However, the purity of this stream is vigorously protected by waste management statutes.

- **20 PAP (Corrugated Fiberboard):** Used for shipping boxes. The graph needs a Mechanical_Instruction attribute: "Must be flattened" (*Flach zusammenfalten*). Failure to flatten causes bin overflow and collection inefficiency.[7]
- **21 PAP (Non-corrugated Fiberboard):** Used for cereal boxes, folding cartons. High-quality fibers.
- **22 PAP (Paper):** Used for bags, newspapers.
- **The "Dirty Paper" Exception:** A pizza box marked "20 PAP" that is heavily soiled with grease moves from the **Blue Bin** node to the **Residual Waste** (*Restmüll*) node. The grease contaminates the pulping process. The graph must model this conditional logic based on the Condition state of the object.[22]

4.2 Metals (40-41)

Metals are unique because their sorting is physical (magnetic) rather than optical (NIR).

- **40 FE (Steel):** Recovered via overband magnets. Found in food cans (*Konservendosen*) and lids.
- **41 ALU (Aluminum):** Recovered via Eddy Current Separators (*Wirbelstromscheider*).

Found in trays, foils, and beverage cans.
- **Disposal:** In most of Germany, these go to the **Yellow Bin**. However, some municipalities (e.g., Munich, parts of Bavaria) use a "Dosencontainer" (Depot system) for metal, keeping it out of the sack system. The graph requires a Location_Override parameter for these codes.[16]

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5. The Composite Ontology: Deciphering the "C/" Codes (80-98)

Composites (*Verbundverpackungen*) represent the most significant source of consumer confusion and data complexity. The 97/129/EC system uses a specific notation: C/ followed by the abbreviation of the *predominant* material by weight.[8]

5.1 The Logic of Composites

The graph must define a composite not just by its code, but by its constituent layers.

- **Entity:** CompositePackaging
- **Attribute 1:** PredominantMaterial (Determines the Abbreviation, e.g., PAP).
- **Attribute 2:** SecondaryMaterial (Determines the Number range).
- **Disposal Rule:** In Germany, *sales packaging* composites generally go to the **Yellow Bin**, even if they look like paper. The "Dual System" pays for the separation of these materials.

5.2 Critical Composite Codes

The following codes are essential for the knowledge graph, as they map to specific recycling technologies or issues.

Table 2: Composite Material Codes and German Processing

Code	Label	Composition (Dominant/Secondary)	Typical Application	German Disposal & Processing
80	**C/PAP**	Paper / Metals	Biscuit canisters with metal lining.	**Yellow Bin.** Hard to recycle; often thermal recovery.
81	**C/PAP**	Paper / Plastic	Bakery bags	**Yellow Bin**

			with windows, frozen food boxes.	(usually). *Exception:* If materials are separable by hand (e.g., peelable window), consumer separates: Paper -> Blue, Plastic -> Yellow.[22]
82	**C/PAP**	Paper / Aluminum	Butter wrappers, cigarette packs.	**Yellow Bin.** Aluminum is recovered; paper fibers often lost or low quality.
84	**C/PAP**	Paper / Plastic / Aluminum	**Liquid Beverage Cartons (FKN):** Tetra Pak, Elopak.	**Yellow Bin.** This is a critical node. Code 84 items are sorted into a dedicated fraction (FKN) and sent to specialized paper mills (ReCarton process) to recover fibers.[16]
90	**C/LDPE**	Plastic / Aluminum	Crisp bags, coffee packs.	**Yellow Bin.** Often difficult to separate; chemical recycling or thermal

				recovery.
91	**C/PET**	Plastic / Tinplate	Blister packs.	**Yellow Bin.**
92	**C/PL**	Plastic / Misc. Metals	Technical packaging.	**Yellow Bin.**
95	**C/GL**	Glass / Plastic	Glass bottles with full-body sleeves.	**Glass Container.** *Warning:* If the sleeve is too thick, it can confuse sorting. ZSVR recommends perforated sleeves to aid separation.

5.3 The "5% Rule" and Grading

According to the ZSVR Minimum Standard, a paper-based packaging with a plastic coating can only be considered "Paper" (for the Blue Bin) if the non-paper component is less than 5% of the total mass.[4]

- **Graph Logic:**
  - Node: Paper_Cup
  - Attribute: Plastic_Content_Percentage
  - Logic: IF Plastic_Content_Percentage > 5% THEN Code = 81 AND Destination = Yellow Bin. ELSE Code = 21 AND Destination = Blue Bin.

This nuances the data significantly. A code "81" explicitly tells the graph "Do not put me in the Blue Bin," preventing contamination of the paper stream.[22]

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6. The German Disposal Logic Layer: The "Abfall-ABC"

A static code means nothing without a dynamic disposal instruction. The knowledge graph must map the codes to the physical infrastructure of German waste management, which varies by region and by the specific "Dual System" operating in the area.

6.1 The Container Nodes (Target Destinations)

The graph must contain nodes representing the physical bins.

1. **Gelber Sack / Gelbe Tonne (Yellow Sack/Bin):**
    - *Input:* Codes 01-07 (Non-deposit), 40-41, 80-98.
    - *Definition:* For "Leichtverpackungen" (LVP) - Lightweight Packaging.
    - *System:* Dual System (Privatized).
    - *Exclusion:* Rigid plastics that are *not* packaging (e.g., a plastic toy) are excluded in jurisdictions that strictly use the "Gelber Sack".[21]
2. **Wertstofftonne (Recycling Bin):**
    - *Input:* Same as Yellow Bin + "Stoffgleiche Nichtverpackungen" (Material-identical non-packaging).
    - *Definition:* A modernized collection system used in cities like Berlin (BSR). It accepts the plastic toy that the Gelber Sack rejects.
    - *Graph Attribute:* Municipality_Has_Wertstofftonne (Boolean). This toggles the destination of non-packaging plastics.[26]
3. **Blaue Tonne (Paper Bin):**
    - *Input:* Codes 20, 21, 22.
    - *Exclusion:* Code 81 (unless <5% plastic), soiled paper (Pizza boxes), receipt paper (thermal paper often contains phenols, belongs in Residual).
4. **Glascontainer (Glass Igloos):**
    - *Input:* Codes 70 (White), 71 (Green), 72 (Brown).
    - *Logic:* Color separation is mandatory. Blue or Red glass must be mapped to the **Green** container (as green glass can absorb the most color impurity).[16]
5. **Restmüll (Residual Waste):**
    - *Input:* Dirty packaging, Code 03 (PVC, if specified by local bylaw), hygiene items, crystal glass, ceramics.

6.2 The Dual System Ecosystem

The recycling of packaging in Germany is financed by the "Dual Systems." Manufacturers pay license fees based on the weight and material type of their packaging. The graph should include an entity type for DualSystem to link to compliance data.

- **Entities:** Der Grüne Punkt, BellandVision, Zentek, Interseroh, etc..[10]
- **Function:** They organize the collection (Yellow Bins) and sorting.
- **Relation:** Packaging_Object -> Licensed_Via -> DualSystem.

6.3 Regional Variations: Berlin vs. Munich

The data sources reveal distinct "dialects" of disposal rules.[21]

- **Berlin (BSR):** heavily promotes the **Wertstofftonne**. Their data (available via Open Data portals) allows for a broad inclusion of plastics.[26]

- **Munich (AWM):** Relies heavily on the **Wertstoffhof** (Recycling Center) system. Residents are encouraged to bring specific fractions to depots rather than just using curbside bins.
- **Graph Implementation:** The graph needs a Region node.
  - Rule_Berlin: Plastic Frying Pan -> Wertstofftonne.
  - Rule_Munich: Plastic Frying Pan -> Wertstoffhof (or Restmüll if depot unavailable).

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7. Digital Data Sources & Ingestion Strategy

To automate the population of this graph, one must tap into existing digital repositories. The research identifies three primary data streams: Open Food Facts, GS1 GDSN, and Wikidata.

7.1 Open Food Facts (OFF) Taxonomies

Open Food Facts maintains a sophisticated, crowdsourced taxonomy of packaging data, stored as hierarchical text files in their GitHub repository.

- **Source:** taxonomies/packaging_materials.txt.[30]
- **Structure:** A Directed Acyclic Graph (DAG).
  - en:plastic is a parent of en:pet.
  - en:pet is a parent of en:rpet (Recycled PET).
- **Ingestion Logic:** The graph should parse the OFF taxonomy to seed the class hierarchy.
  - OFF Tag en:tetra-brik -> Mapped to Graph Code 84.
  - OFF Tag en:glass -> Mapped to Graph Class Glass (Codes 70-72).
  - OFF Tag en:plastic-tray -> Mapped to Form: Tray -> Triggers Yellow Bin logic.

7.2 GS1 Global Data Synchronization Network (GDSN)

For B2B integration, the GS1 standards provide the packagingMaterialTypeCode. This is a controlled vocabulary used by retailers and manufacturers.[19]

- **Relevance:** When a German retailer lists a product, they receive GS1 data. The graph acts as a translator between this B2B data and the consumer-facing 97/129/EC codes.
- **Mapping:**
  - POLYMER_PET -> 01.
  - PAPER_CORRUGATED -> 20.
  - LAMINATED_CARTON -> 81 or 84 (Ambiguous; requires secondary attribute check).

7.3 Wikidata Properties

Wikidata offers a semantic web-ready source of identifiers.

- **Property: P5994** (Recycling Code).[33]
- **Usage:** The graph can query Wikidata to retrieve the official recycling code for a given material entity (e.g., Q144256 for Polyethylene Terephthalate). This links the graph to the

broader semantic web, allowing access to multilingual labels and chemical formulas.[34]

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8. Assessing Recyclability: The ZSVR Minimum Standard

A modern waste management graph cannot simply stop at "Identity." It must assess "Recyclability." The ZSVR *Mindeststandard* defines the algorithm for this assessment.[4]

8.1 The Three-Step Verification

The knowledge graph should implement a verification function verifyRecyclability(PackagingItem) based on ZSVR rules:

1. **Existence of Infrastructure:** Is there a collection and sorting structure for this material?
   - *True:* PET, PE, PP, Paper, Steel, Aluminum, Glass.
   - *False/Limited:* Bioplastics (PLA), PVC packaging, Polystyrene (EPS) in Yellow Bin.[22]
2. **Sortability (NIR Detection):** Can the sorting facility identify the item?
   - *Constraint:* **Carbon Black**. Standard black pigments absorb NIR radiation, making the plastic invisible to sorters. It ends up in residue.
   - *Data Point:* If Color = Black AND Pigment!= NIR-Reflective, THEN Recyclable = False.[12]
3. **Incompatibility (Density/Wash):** Does the item disrupt the recycling of the target fraction?
   - *Example:* A PET bottle (Density 1.38) with a PVC full-body sleeve (Density 1.35). Both sink in the wash tank. The PVC contaminates the PET flakes.
   - *Graph Rule:* If Body = PET AND Component = PVC, THEN Recyclability_Grade = Downgraded or Fail.

8.2 Design for Recycling (D4R) Guidelines

The graph should incorporate data from **RecyClass** and **Cyclos-HTP**.[36] These organizations publish tables of "Compatibility."

- **Green:** Fully compatible (e.g., Clear PET bottle + PP cap).
- **Yellow:** Limited compatibility (e.g., PET bottle + heavily printed sleeve).
- **Red:** Incompatible (e.g., PET bottle + metal cap that doesn't detach).

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9. Waste Codes (AVV): The Industrial Interface

While consumers look for the "Green Dot" or the Code 01, waste management companies operate using the **Abfallverzeichnis-Verordnung (AVV)** codes.[38] The knowledge graph must map the consumer-facing reality to this industrial accounting system.

Table 3: Mapping Consumer Codes to AVV Industrial Codes

Consumer Code	Material	AVV Code	AVV Description
20, 21, 22	Paper/Cardboard	**15 01 01**	*Verpackungen aus Papier und Pappe*
01 - 07	Plastics	**15 01 02**	*Verpackungen aus Kunststoff*
50, 51	Wood	**15 01 03**	*Verpackungen aus Holz*
40, 41	Metals	**15 01 04**	*Verpackungen aus Metall*
80 - 98	Composites	**15 01 05**	*Verbundverpackungen*
Mixed	Unsorted	**15 01 06**	*Gemischte Verpackungen*
70 - 72	Glass	**15 01 07**	*Verpackungen aus Glas*

Operational Insight:
Code 15 01 05 (Composites) is a friction point. In a sorting plant, a "Code 81" (Paper/Plastic) might be processed. If the plastic is <5%, the output is marketed as 15 01 01 (Paper). If it is >5%, it remains 15 01 05 or moves to energy recovery. The graph must distinguish between the input AVV (collection) and the output AVV (post-sorting).40

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10. Conclusion: The Knowledge Graph as a Semantic Bridge

The creation of a German Waste Management Knowledge Graph is an exercise in translating between three languages: the legal language of the EU (97/129/EC), the technical language of the polymer industry (GS1/DIN), and the pragmatic language of the consumer (Abfall-ABC).

This research report establishes that a linear mapping (Code -> Bin) is insufficient. The graph must handle:

1. **Conditional Disposal:** Driven by regional infrastructure (Wertstofftonne vs. Gelber Sack) and packaging type (Sales vs. Transport).
2. **Recyclability Logic:** Driven by physical properties (Density, NIR detectability) rather than just material identity.
3. **Composite Complexity:** Requiring a decomposition of "C/" codes into their constituent mass percentages.

By populating the graph with the data dictionaries provided in Sections 3, 4, and 5, and implementing the logic layers defined in Sections 6 and 8, the system will accurately reflect the complex reality of the German *Kreislaufwirtschaft*. This infrastructure will support not just compliance for manufacturers (VerpackG), but also clarity for consumers, ultimately driving higher collection rates and purer recycling fractions.

Citations:

- **Legal:**.[1]
- **Standards:**.[4]
- **Material Codes:**.[7]
- **Disposal Rules:**.[16]
- **Data Sources:**.[19]
- **Recyclability:**.[12]

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